

# The G“air”y

Lung Spirometer



Lily Duong, Timmy Nguyen, Ryu Suzuki



# Clinical Motivation

**Chronic  
Obstructive  
Pulmonary Disease  
(COPD)**

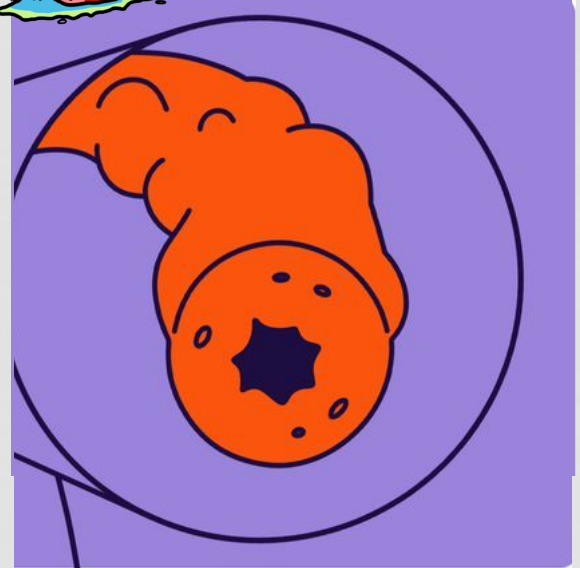


“... is the fourth leading cause of death worldwide, causing 3.5 million deaths in 2021, approximately 5% of all global deaths.” (WHO)

Asthma affected an estimated 363 million people in 2023 (IHME)



**Asthma**



# Clinical Role

Doctors order spirometry if patients exhibiting symptoms, also for monitoring.

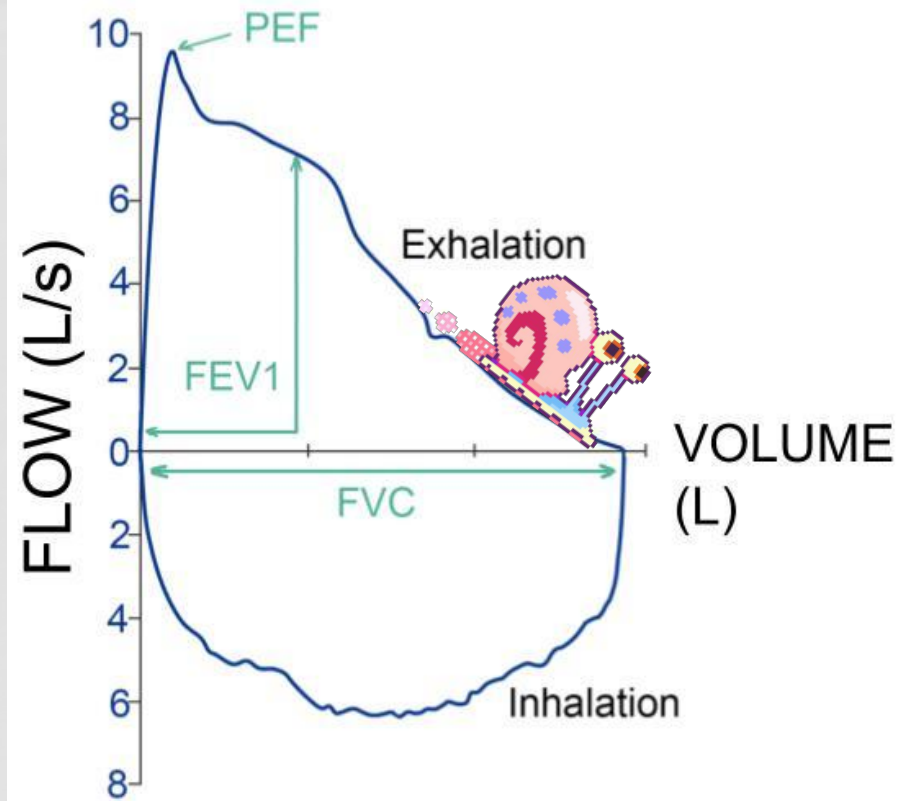
Measurements: airflow  
& lung volume (15-25 Hz)

**FEV<sub>1</sub>**: air exhaled in  
1 second (L)

**FVC**: total exhaled air (L)

**Screening ratio:**

**FEV<sub>1</sub>/FVC < 0.7**  
-> obstructive disease.



➡ Initiate treatment plans



# Physiologic Requirements → Engineering Design

**Breathing frequencies:** up to  
~15–25 Hz

**BW:** ~30 Hz, set AAF to 40 Hz

**Airflow:** up to 10 L/s in  
forced expiration

**Sampling Rate:** >200 Hz

**Measurement duration:**  
~2-5 seconds

**Power/Weight:** battery powered,  
portable

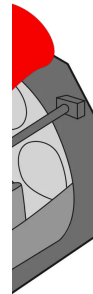


# Instruments to measure flow rate/volume

$Q \rightarrow$  (



ph  $\rightarrow$  Best  
e of  
 $\rightarrow$  linear  
 $\rightarrow$  robust  $\Delta P$



Roll  
mea  
air p



# That's why we use the **VENTURI** **TUBE** (it's cheap/easy to make)



We exploit the  
Venturi effect:

Lower pressure  
higher flow

Higher pressure  
lower flow

To get a pressure  
difference!



# Overarching Post-Proce

BERNOULLI'S E  
velocity changes  
eventually volum

$$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

**REARRANGE!**

$$p_1 + \frac{1}{2}\rho$$

$$\Delta P = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

Assume incompressible fluid, mass flow rate is constant  
(conservation of mass)

## Continuity equation

$$A_1 V_1 = A_2 V_2$$

$$v_1 = \frac{A_2}{A_1} v_2$$

$$v_1^2 = \left( \frac{A_2}{A_1} \right)^2 v_2^2$$

$$\Delta P = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

Solve for  $v^2$  and simplify

$$v_2^2 = \frac{2\Delta P}{\rho(1 - (A_2/A_1)^2)}$$

$$v_2 = \sqrt{\frac{2\Delta P}{\rho(1 - (A_2/A_1)^2)}}$$

We know that  
 $Q_2 = A_2 \cdot v_2$

$$Q = A_2 \sqrt{\frac{2\Delta P}{\rho(1 - (A_2/A_1)^2)}}$$

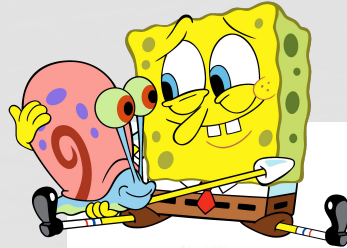
Integrate over time to get Volume  
( $0 \rightarrow 1s$  for FEV1, and  $0 \rightarrow$  total  
breathing time for FVC)

$$V(t) = \int_0^t Q(\tau) d\tau$$



```
if pressure_pa == 0:  
    flow_m3s = 0.0  
else:  
    flow_mag = A2 * math.sqrt(  
        (2.0 * abs(pressure_pa)) /  
        (rho * (1.0 - (A2 / A1) ** 2))  
    )  
    flow_m3s = math.copysign(flow_mag, pressure_pa)
```

## Post-Processing for Flow Rate



$$Q = A_2 \sqrt{\frac{2\Delta P}{\rho (1 - (A_2/A_1)^2)}}$$

```
fev1_end = min(start_idx + int(FS * 1.0), end_idx)

if fev1_end > start_idx:
    fev1 = sum(abs(flows_Ls[i]) * DT for i in range(start_idx, fev1_end))

if end_idx > start_idx:
    fvc = sum(abs(flows_Ls[i]) * DT for i in range(start_idx, end_idx))

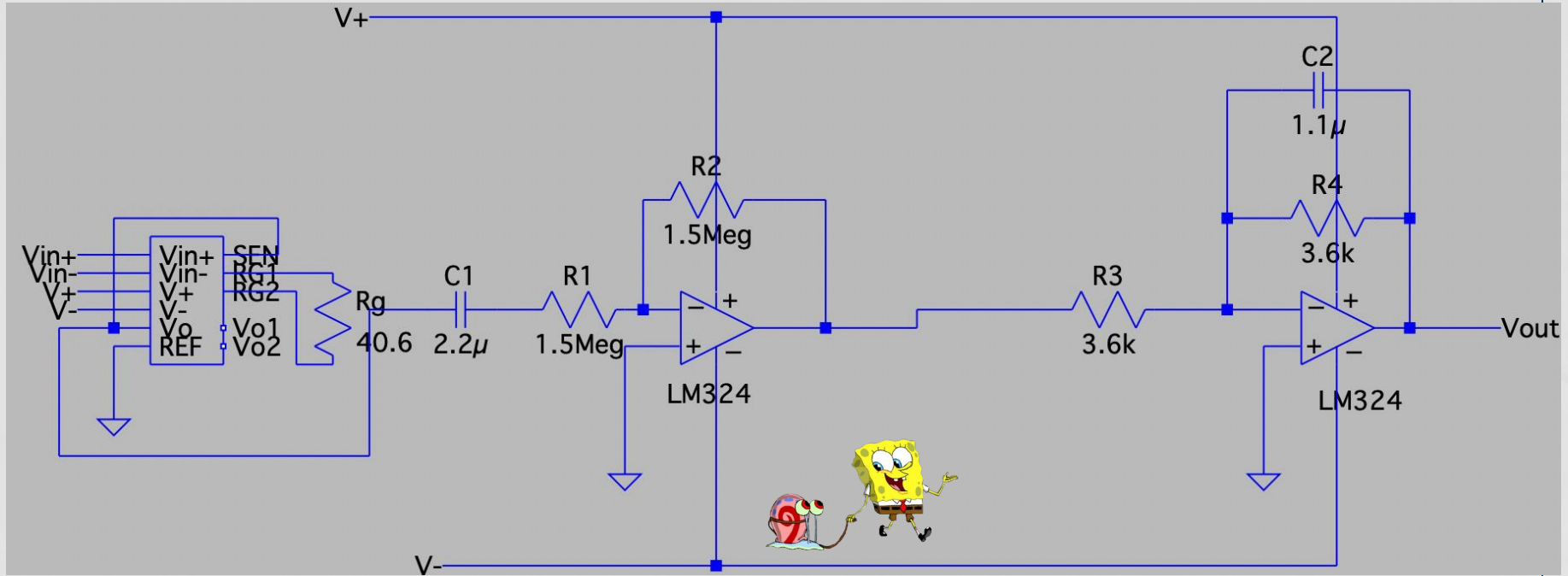
print("start_idx:", start_idx, "end_idx:", end_idx, "duration_s:", (end_idx - start_idx) * DT)
```


$$V(t) = \int_0^t Q(\tau) d\tau$$

Post-  
Processing  
for  
Volume  
(discrete  
sum)

# LTSPICE Circuit

## Band pass filter

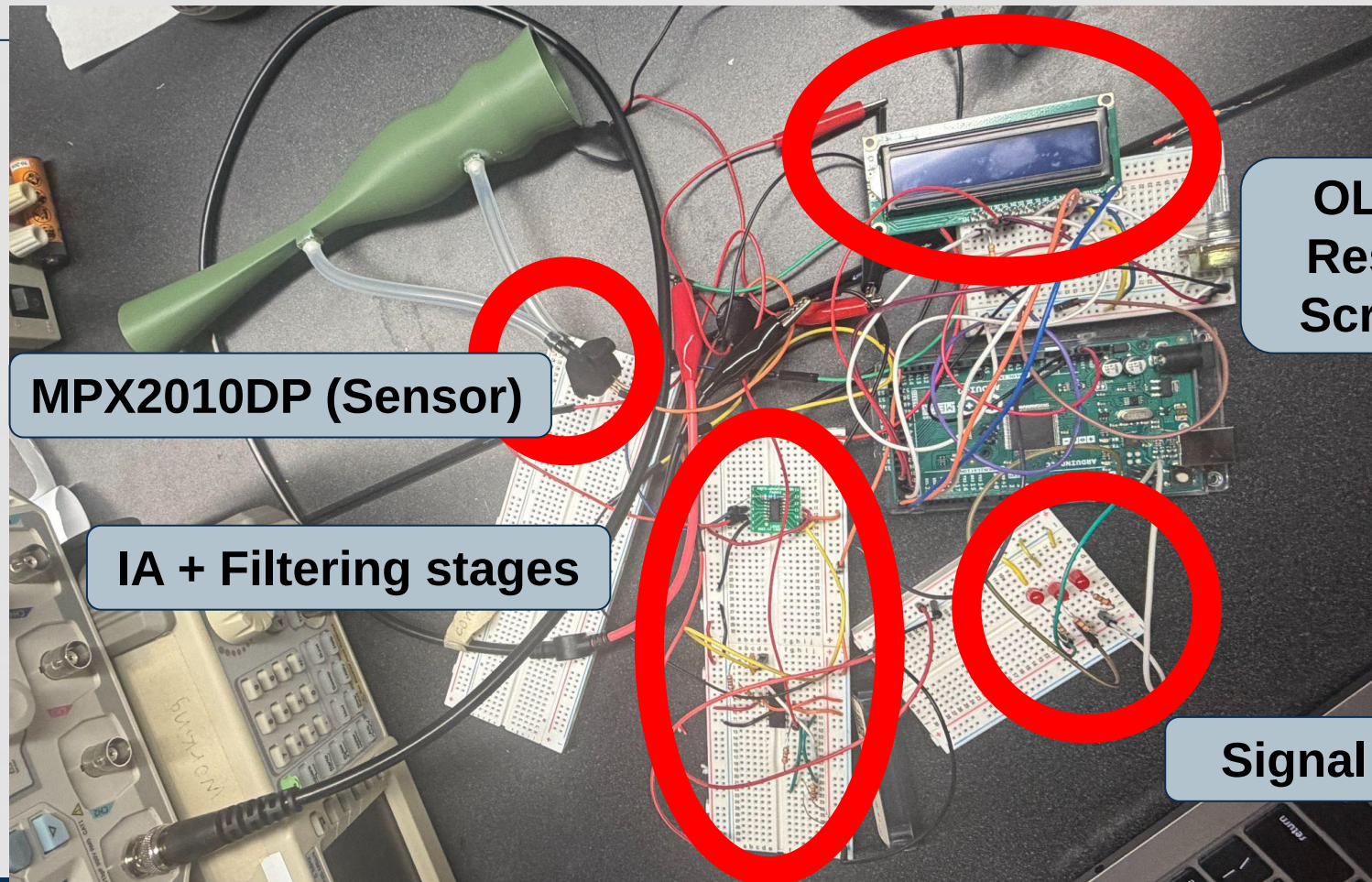


Preamp (INA163)

High-pass filter (0.05 Hz)

Low-pass filter (40 Hz)





**MPX2010DP (Sensor)**

**IA + Filtering stages**

**OLED  
Result  
Screen**

**Signal LEDs**



# Sensor Specs

## Differential Pressure Sensor

**MPX2010DP**

**Sensitivity:** 2.5 mV/kPa

**Operating pressure range:** 0-10 kPa

**Source impedance:**

- Input: 1300-2550  $\Omega$
- Output: 1400-3000  $\Omega$



Assume  $R_s = 2000$

**BW:** 349.8 Hz (based on response time from 10% - 90%)



# Sensor Signal and Noise and Interference Model

**INA163 (Gain =  $1 + 6000/R_g$ )**

$$\begin{array}{ll} E_n = 1 \text{ nV} / \sqrt{\text{Hz}} & R_n = 1250 \\ I_n = 0.8 \text{ pA} / \sqrt{\text{Hz}} & P_n = 8 \cdot 10^{-22} \text{ W/Hz} \end{array} \rightarrow$$

$$R_s = [1400, 3000] \Omega$$

**Optimal Noise Matching (assume  $T=300\text{K}$ )**

$$F = [1.097, 1.136]$$



$$F = \frac{\overline{N_{\text{total}}^2}}{N_{\text{thermal}}^2} = \frac{[4kTR_s + e_n^2 + i_n^2 R_s^2] \Delta f}{4kTR_s \Delta f} = \left[ 1 + \frac{e_n^2}{4kTR_s} + \frac{i_n^2 R_s}{4kT} \right]$$



# Sensor Signal and Noise and Interference Model

SNR

$$\text{SNR}_{\text{overall}} = \frac{S}{\sigma_{N_{\text{total}}}} = \frac{S}{\sqrt{1.1} \sigma_{N_{\text{thermal}}}} = \frac{\text{SNR}_{\text{thermal}}}{\sqrt{1.1}}$$

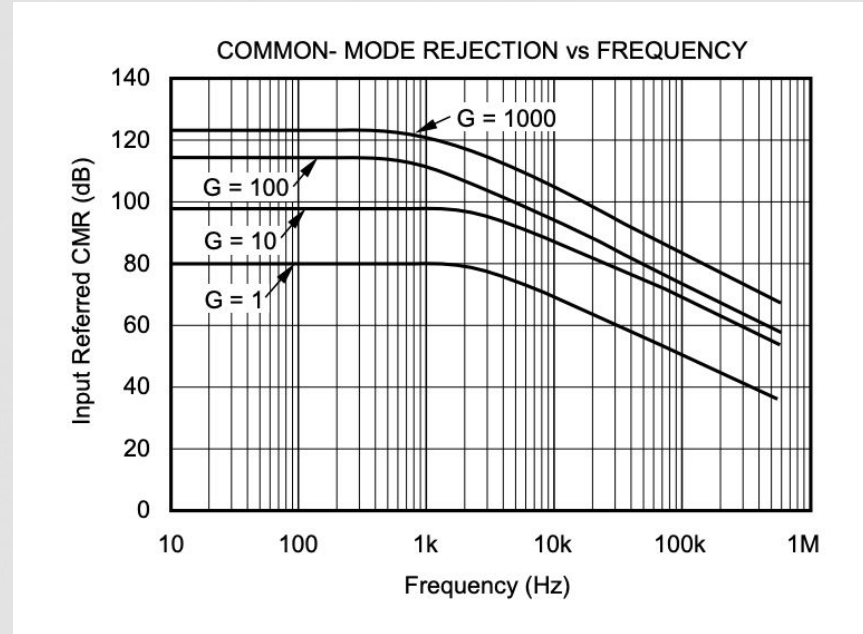


## CMRR? HOW MUCH WILL 60Hz be rejected

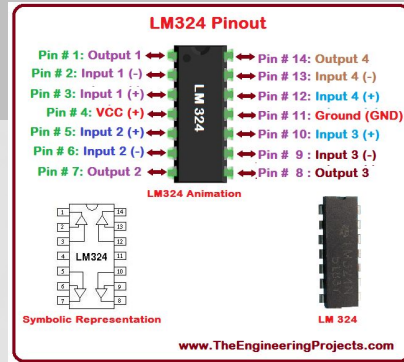
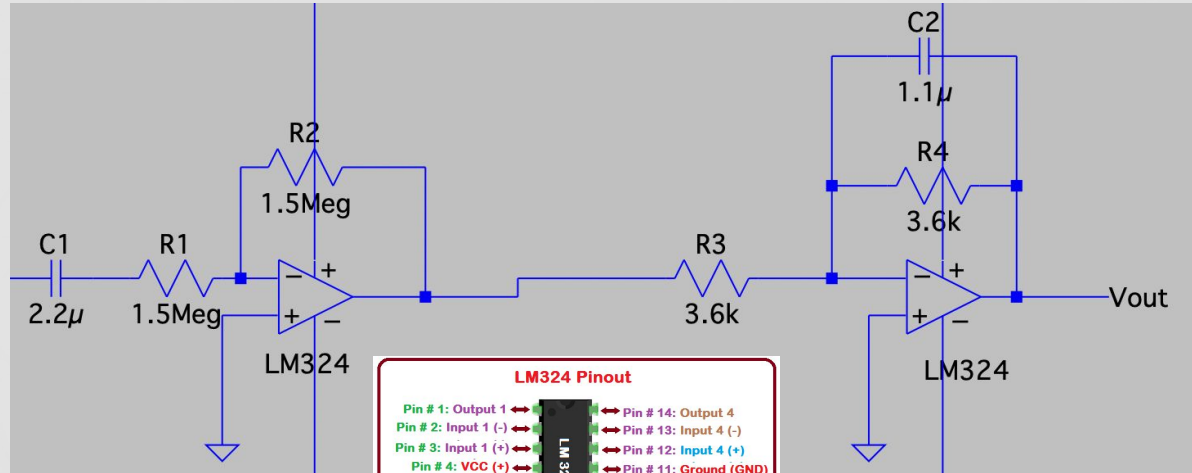
Our Gain?  $R_g = 40.6$

Gain = 148

- CMR ? ~ 105
- $V_{\text{common mode}}$  attenuated by  $8 \times 10^{-4}$
- We will have further attenuation of 60Hz through filtering too!



# Filtering



**Band Pass filter**  
**LM324**

**0.5 Hz HPF**

**40 Hz LPF:**  
- Also our AAF



# AAF and ADC

## Arduino Mega

$F_c = 40\text{Hz}$  ( set by 3rd stage filter).

Sensor SNR = 55  $\rightarrow$  need 10 bits (Prudent Rule)



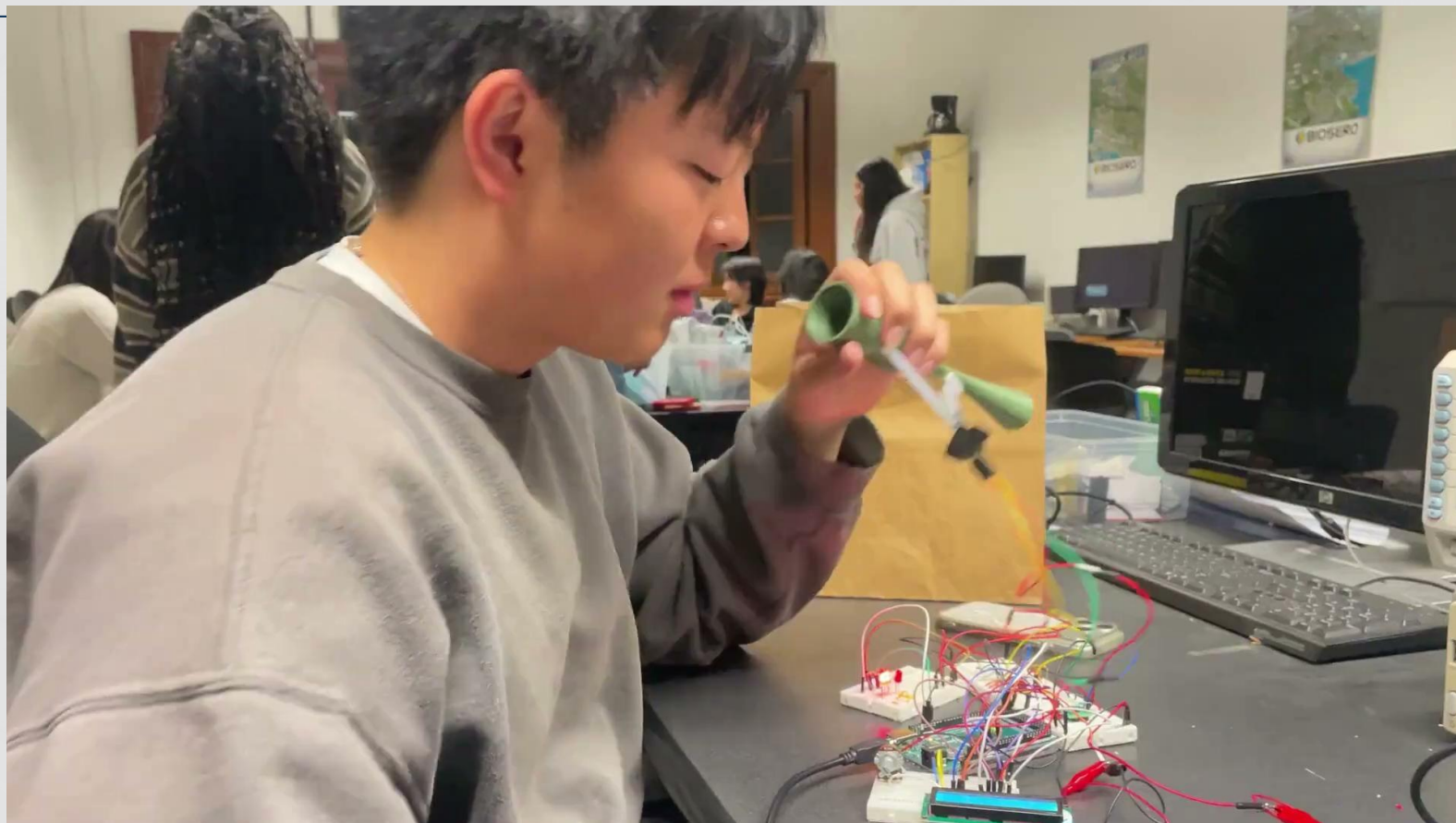
**ADC**  $F_c = 40\text{Hz}$   
( set by 3rd stage filter).

**Op Amps** LM324

$f_s > 93.33\text{ Hz} \rightarrow$  sampled at  $100\text{Hz}$  to at least  
attenuate the noise aliasing to 3 order.



$$f_s > BW(1 + A^{1/N}) \text{ for signal ranging over } f = \pm BW \text{ (Prudent Nyquist sampling rate)}$$



# FEV1 (L) Comparison

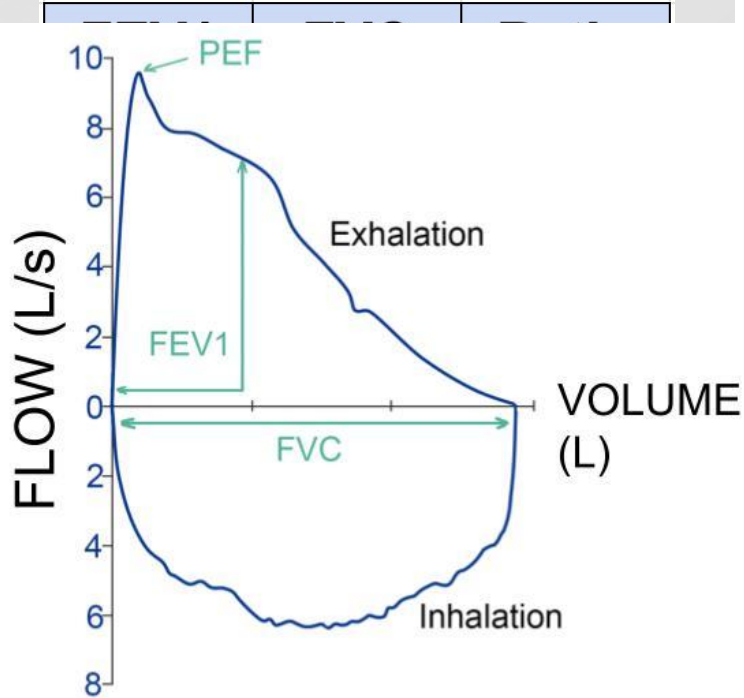


| Gold Standard | Gairy | % Error |
|---------------|-------|---------|
| 3.12          | 3.19  | -2.24   |
| 2.89          | 2.48  | 14.19   |
| 3.31          | 3.45  | -4.23   |
| 3.34          | 3.53  | -5.69   |
| 3.46          | 3.49  | -0.87   |
| 3.45          | 3.33  | 3.48    |
| 3.54          | 2.92  | 17.51   |
| 3.35          | 3.17  | 5.37    |
| 3.4           | 3.42  | -0.59   |
| 3.3           | 3.44  | -4.24   |

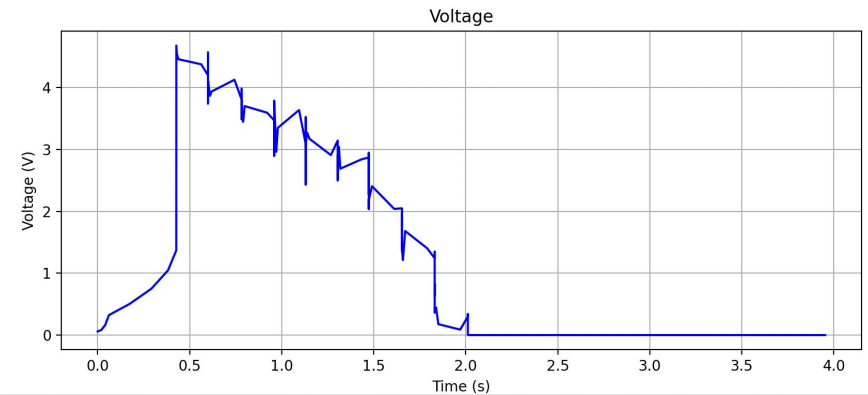
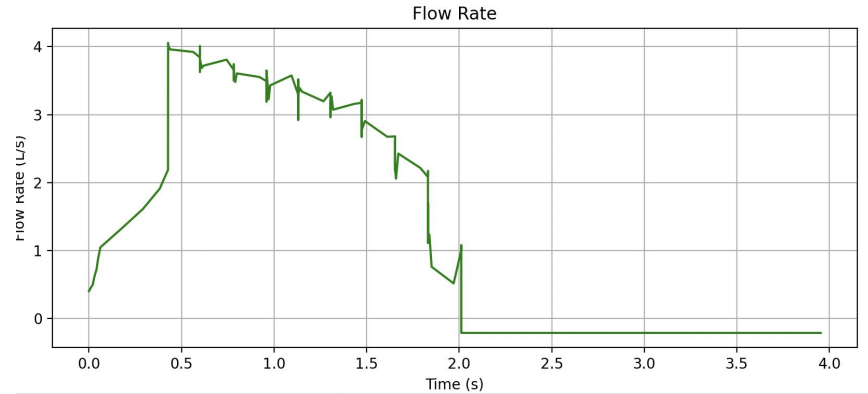


**Average ~ 5.8% error**

# FEV1/FVC Results



|      |      |      |   |
|------|------|------|---|
| 3.42 | 4.01 | 0.85 | ✓ |
| 3.44 | 4.62 | 0.74 | ✓ |





# FEV1/FVC

| Avg FEV1 | Avg FVC |
|----------|---------|
| 3.242    | 4.56    |

HEALTHY!



| FEV1 | FVC  | Ratio |
|------|------|-------|
| 3.19 | 5.12 | 0.62  |
| 2.48 | 4.03 | 0.62  |
| 3.45 | 4.21 | 0.82  |
| 3.53 | 4.94 | 0.71  |
| 3.49 | 4.91 | 0.71  |
| 3.33 | 5.7  | 0.58  |
| 2.92 | 4.67 | 0.63  |
| 3.17 | 3.48 | 0.91  |
| 3.42 | 4.01 | 0.85  |
| 3.44 | 4.62 | 0.74  |

# Trade Offs



## Venturi implementation

- ✓ Accurate physics model (Bernoulli-based flow)
- ✗ 3D printing → rough surface & imperfect dimensions.

## “Cheap” Sensor

Chose sensor with low sensitivity (2.5 mV/kPa)  
→ Low-cost components  
→ Lower precision than clinical devices.



# Lessons learned

**Debugging sucks!!**

Gain vs. Saturation



Are batteries plugged in??

We can read data sheet now?





# Final Takeaways

## **Built a working low-cost spirometer:**

- Successfully measured airflow using pressure sensing
- Converted signal → clinically relevant metrics (FEV<sub>1</sub>, FVC, FEV<sub>1</sub>/FVC)

# Thanks!



Us yesterday



# Works Cited

1. [https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-\(copd\)#:~:text=Key%20facts,risk%20of%20other%20health%20problems](https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-(copd)#:~:text=Key%20facts,risk%20of%20other%20health%20problems)
2. <https://www.healthdata.org/research-analysis/library/global-burden-disease-2023-findings-gbd-2023-study>
3. <https://www.nxp.com/docs/en/data-sheet/MPX2010.pdf>
4. [https://www.ti.com/product/INA163?utm\\_source=google&utm\\_medium=cpc&utm\\_campaign=asc-null-null-GPN\\_EN-cpc-pf-google-ww\\_en\\_cons&utm\\_content=INA163&ds\\_k=INA163&DCM=yes&gclsrc=aw.ds&gad\\_source=1&gad\\_campaignid=14388345080&gbraid=0AAAAAC068F1-ssym411m5XT8CqYo22Yhk&gclid=CjwKCAjwzLHPBhBTEiwABaLsSnUnmFjT19ZRFueJiUf\\_sPjeZfKePY\\_W9oZFDz2M5FuCUAlIY4hpxBoCVZcQAvD\\_BwE](https://www.ti.com/product/INA163?utm_source=google&utm_medium=cpc&utm_campaign=asc-null-null-GPN_EN-cpc-pf-google-ww_en_cons&utm_content=INA163&ds_k=INA163&DCM=yes&gclsrc=aw.ds&gad_source=1&gad_campaignid=14388345080&gbraid=0AAAAAC068F1-ssym411m5XT8CqYo22Yhk&gclid=CjwKCAjwzLHPBhBTEiwABaLsSnUnmFjT19ZRFueJiUf_sPjeZfKePY_W9oZFDz2M5FuCUAlIY4hpxBoCVZcQAvD_BwE)
- 5.

